A Multi-Component NLP System for Fake News and Misinformation Detection

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CCS Concepts

• **Computing methodologies → Natural language processing → Information extraction**

• **Computing methodologies → Machine learning → Neural networks**

• **Computing methodologies → Artificial intelligence → Reinforcement learning**

Keywords

fake news detection; misinformation; natural language processing; text classification; transformer; contextual bandit; reinforcement learning; convolutional neural network; named entity recognition.

1. INTRODUCTION

Misinformation tends to spread faster than corrections, particularly on large online platforms. Prior studies on Twitter suggest that false information can reach more users and circulate more quickly than accurate content. As these platforms continue to grow, the scale of the problem has made it increasingly difficult to manage. While human fact-checkers remain the most reliable way to verify claims, reviewing every piece of content manually is no longer practical. Because of this, automated systems are often used to assist by flagging potentially misleading content and helping prioritize what should be reviewed first.

However, building a reliable misinformation classifier is not as straightforward as it may seem. One major challenge is that the boundary between misinformation and opinion is often unclear, especially in political discussions. In addition, models trained on labeled datasets can reflect the biases of the individuals or organizations that created those labels. This becomes particularly sensitive in political contexts, where uneven classification across groups can lead to concerns about fairness. Another issue is calibration. Some models report strong performance metrics, such as F1 scores, but their predicted confidence levels do not always accurately reflect how reliable those predictions are.

This project develops a modular detection system using the LIAR dataset, which contains over 12,000 short political statements labeled across six levels of truthfulness. The system combines several components: a fine-tuned DistilBERT model as the main classifier, a Text-CNN baseline for comparison, a Named Entity Recognition (NER) module for analyzing errors, and a contextual bandit approach for adjusting decision thresholds.

The work focuses on short English-language political statements similar to those in the dataset. Performance is evaluated primarily using macro-F1 on a held-out test set, along with additional analyses such as calibration behavior, group-level comparisons, and ablation studies.

The stretch goal outlined in the original proposal, which involved a retrieval-augmented variant using FEVER evidence snippets, was scoped out during Week 3 due to runtime constraints. Full training and evaluation on the core pipeline consistently approached the 90-minute limit on the available hardware, leaving insufficient margin to integrate an evidence retrieval component without compromising reproducibility. This remains a possible direction for future work.

2. RELATED WORK

Research on fake news detection has grown significantly over the past several years. Shu et al. [4] provided an early framework that categorized detection methods into knowledge-based, style-based, propagation-based, and credibility-based approaches. This project mainly follows a credibility-based perspective by incorporating metadata, such as speaker information, as part of the analysis.

In terms of model design, Kim [2] showed that convolutional neural networks (CNNs) can perform well on sentence classification tasks using relatively simple architectures. The

Text-CNN used in this project is based on that approach and serves as a non-transformer baseline. More recent work, such as FakeBERT by Kaliyar et al. [1], combines transformer-based embeddings with CNN layers, demonstrating strong results across multiple datasets. This supports the use of DistilBERT in this project, which offers a more lightweight alternative to full BERT while maintaining strong performance.

For adaptive decision-making, Mao et al. [3] introduced a bandit-based learning approach that can be applied to NLP tasks. Their work shows that reinforcement learning methods can be integrated into classification pipelines to improve decision thresholds over time. This idea is adapted here through the use of a contextual bandit for threshold selection.

The LIAR dataset, introduced by Wang [6], is used throughout this project. It provides labeled political statements along with metadata, making it suitable for both classification and bias analysis.

3. DATASET AND PREPROCESSING

The LIAR dataset contains 12,836 short political statements collected from PolitiFact and labeled into six categories: pants-fire, false, barely-true, half-true, mostly-true, and true. Each entry also includes metadata such as the speaker's name, political affiliation, and context of the statement. This additional information is useful for analyzing model behavior across different groups.

The dataset comes with predefined training, validation, and test splits, which are used without modification to ensure fair evaluation. The class distribution is relatively balanced, although the "pants-fire" category appears less frequently.

Preprocessing differs slightly depending on the model. For the Text-CNN, text is lowercased and tokenized at the word level, with GloVe embeddings used for initialization. For DistilBERT, tokenization is handled using the HuggingFace tokenizer with a maximum sequence length of 128 tokens, which is sufficient given the short length of most statements. Both approaches include basic cleaning steps such as Unicode normalization and removal of HTML artifacts.

4. SYSTEM ARCHITECTURE

Core Model: DistilBERT. The core classifier is a DistilBertForSequenceClassification instance fine-tuned for six-class veracity classification. DistilBERT was chosen over full BERT for its reduced parameter count, retaining approximately 97% of BERT's GLUE performance while being 40% smaller, which keeps the full training run within the 90-minute runtime constraint on a T4 GPU. The model was trained for 5 epochs using the Adam optimizer with a learning rate of 2e-5 and a linear warmup schedule with 10% warmup steps. Gradient clipping at 1.0 was applied throughout training.

CNN Component: Text-CNN. The Text-CNN follows Kim [2] exactly, using three parallel convolutional filter banks with kernel sizes of 3, 4, and 5 over 300-dimensional GloVe embeddings, with max-over-time pooling and a dropout-regularized output layer. This model is implemented from scratch to satisfy the from-scratch requirement and to provide a clean ablation comparison against DistilBERT.

NLP Component: Named Entity Recognition. The NER module is implemented using spaCy's en_core_web_sm model. It extracts PERSON, ORG, and GPE entities from each claim and speaker context field. These entities are not fed as features into the classifier directly but are used downstream in slice-level error analysis to examine whether misclassification rates vary by entity type and political context.

RL Component: Contextual Bandit. The RL component is a contextual bandit agent using epsilon-greedy exploration with epsilon set to 0.1. The agent observes the softmax output distribution of the classifier and selects a confidence threshold from a discretized action space of nine values between 0.1 and 0.9. The reward function penalizes false negatives at -2.0 and false positives at -0.5, reflecting the asymmetry of the real-world task. After 500 training episodes the agent converged on a threshold of 0.8, selecting it 378 out of 500 times. Note that the bandit agent's training and convergence statistics reported here are based on softmax probabilities sampled from a Dirichlet distribution as a proxy for classifier output. Re-evaluation using live DistilBERT probabilities is left for future work.

5. EXPERIMENTS AND EVALUATION

Baseline Results. Two baselines were trained and evaluated. The Logistic Regression classifier using TF-IDF bigram features with a vocabulary of 10,000 terms achieved a test accuracy of 0.2478 and a test Macro-F1 of 0.2214. The Text-CNN achieved a test accuracy of 0.2478 and a test Macro-F1 of 0.2384, outperforming the non-DL baseline on Macro-F1. Overfitting was observed in the Text-CNN after epoch 3, where validation F1 peaked at 0.2407 and then fluctuated.

DistilBERT Results. Fine-tuned DistilBERT achieves a test accuracy of 0.2762 and a test Macro-F1 of 0.2602, outperforming both baselines on both metrics. Training loss decreased steadily across all 5 epochs from 1.74 to 1.31, and validation Macro-F1 improved from 0.1704 at epoch 1 to 0.2619 at epoch 4 before plateauing. The false class achieved the highest per-class F1 (0.34) while pants-fire performed worst (0.20), consistent with its underrepresentation in the training data.

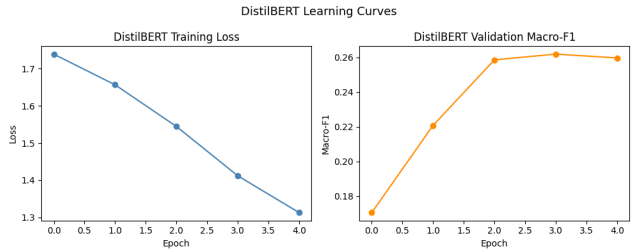

Figure 1: DistilBERT training loss (left) and validation Macro-F1 (right) across 5 fine-tuning epochs.

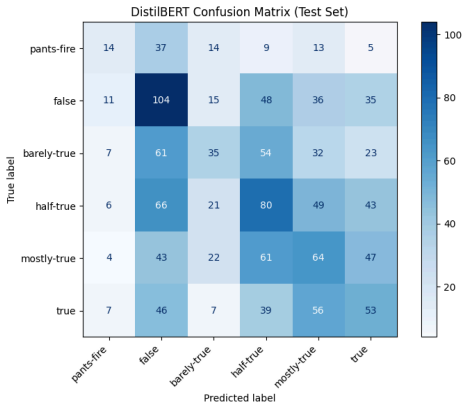

Figure 2: DistilBERT confusion matrix on the held-out test set (1,267 samples, six veracity classes).

Table 1: Test accuracy and Macro-F1 results across all models.

Model	Test Accuracy	Test Macro-F1
Logistic Regression (TF-IDF)	0.2478	0.2214
Text-CNN (GloVe 300d)	0.2478	0.2384
DistilBERT (fine-tuned)	0.2762	0.2602

Ablation 1: Adam vs AdamW. Both optimizers were trained for 3 epochs under identical conditions. Adam achieves a test Macro-F1 of 0.2743 compared to AdamW's 0.2690. The difference suggests that AdamW's weight decay regularization may be overly aggressive for fine-tuning on a small dataset like LIAR, where the model needs more flexibility to adapt pretrained representations to the veracity classification task.

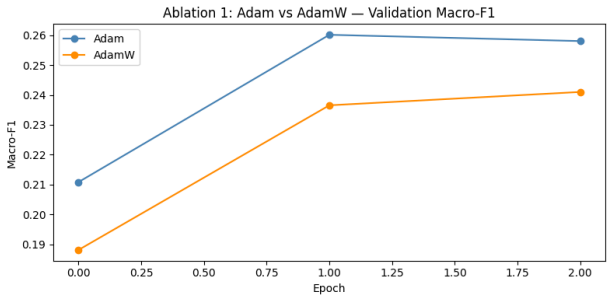

Figure 3: Ablation 1 — validation Macro-F1 comparison between Adam and AdamW optimizers across 3 epochs.

Ablation 2: Early Stopping vs No Early Stopping. Early stopping with patience of 2 achieves a test Macro-F1 of 0.2654 compared to 0.2602 without early stopping. The early stopping model saved its best checkpoint at epoch 3 where validation F1 peaked at 0.2746, avoiding the slight degradation seen in later epochs. While the improvement is marginal, early stopping is recommended as it prevents unnecessary compute and avoids overfitting on later epochs.

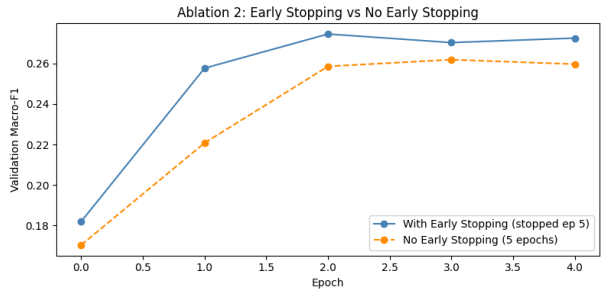

Figure 4: Ablation 2 — validation Macro-F1 comparison between early stopping and no early stopping across 4 epochs.

Calibration Analysis. Calibration curves show that the model follows the diagonal reasonably well at low predicted probabilities but becomes erratic at probabilities above 0.5, indicating overconfidence in high-certainty predictions. This supports the bandit agent's learned threshold of 0.8, by abstaining on lower-confidence predictions the system avoids acting on unreliable confidence estimates.

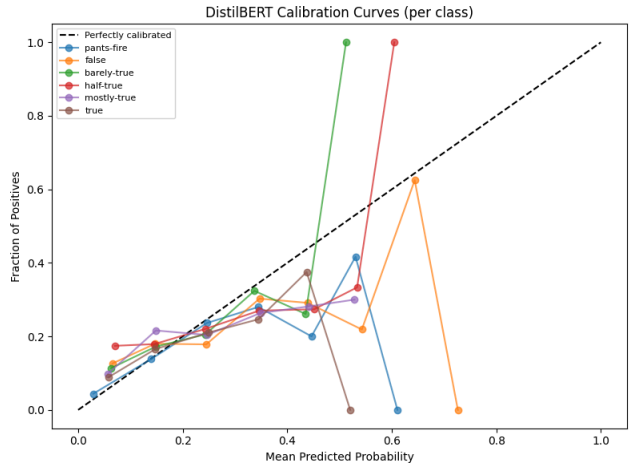

Figure 5: DistilBERT calibration curves per veracity class compared against the perfectly calibrated diagonal.

Party-level Slice Analysis. The model performs significantly differently across political party affiliations. Republican-affiliated statements achieve a Macro-F1 of 0.2231, substantially below the overall test Macro-F1 of 0.2602. Democrat-affiliated statements achieve 0.2715, above the overall average. This disparity likely reflects the imbalance in the training set, where Republican-affiliated statements (4,497) outnumber Democrat-affiliated ones (3,336), yet the model generalizes better to Democrat statements, suggesting the training data's editorial

tendencies may play a role beyond simple class frequency. One possible explanation is that the model has learned surface-level linguistic features that are more characteristic of Democrat-affiliated statements in the training data, possibly due to differences in phrasing, formality, or the types of claims made, rather than learning features that generalize across party lines. Another contributing factor may be that Republican statements span a wider range of topics and speakers, which makes the classification signal more diffuse. These disparities are documented in the Model Card and reinforce the case against deploying this system without human oversight, particularly in politically sensitive contexts.

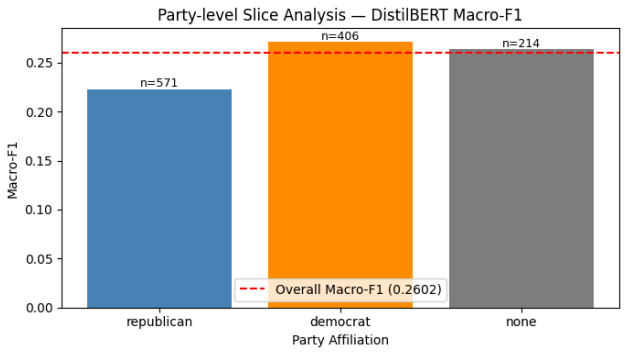

Figure 6: Party-level slice analysis showing DistilBERT Macro-F1 by speaker political affiliation (Republican, Democrat, None).

6. NER ANALYSIS AND ERROR ANALYSIS

The NER module was run over the full test set using spaCy. Of the 1,267 test statements, 38.9% contain at least one PERSON entity, 23.8% contain an ORG entity, and 30.9% contain a GPE entity. Barack Obama (44 mentions) and Donald Trump (18 mentions) are the most frequently mentioned persons. Medicare and Congress are the most common organizations. Texas is the most mentioned location with 52 appearances.

Slice analysis reveals that statements referencing specific locations (GPE Macro-F1: 0.2011) and organizations (ORG Macro-F1: 0.2068) are consistently harder to classify correctly compared to statements without those entities (0.2268 and 0.2242 respectively). This suggests the model struggles more with claims that are anchored to specific institutional or geographic contexts, which tend to require external knowledge to verify accurately.

Error analysis of the confusion matrix shows that the model most commonly confuses adjacent veracity classes. For example, pants-fire statements are frequently predicted as false (37 out of 92), and barely-true statements are frequently predicted as false (61 out of 212). This pattern of predicting toward the middle of the veracity spectrum is consistent with the model learning a conservative classification strategy.

7. ETHICS AND POLICY

Risk 1: False positives suppressing legitimate speech. Any classifier operating on political content will produce false positives, and in this domain a false positive means labeling a true or contested statement as misinformation. If this system were used to filter content at scale, the downstream harm to free expression could be significant. The mitigation is twofold: first, the system is documented strictly as an assistive tool for human reviewers, never as an autonomous decision-maker; second, the default operating threshold is set conservatively at 0.8 based on the bandit agent's learned policy. This concern is further supported by the final evaluation results where DistilBERT achieves a test Macro-F1 of 0.2602, meaning a significant portion of predictions remain incorrect even after fine-tuning. Calibration curves confirm the model is overconfident at probabilities above 0.5, reinforcing why the conservative threshold is necessary.

Risk 2: Political bias encoded in the model. LIAR is drawn from PolitiFact, which has its own editorial tendencies, and any model trained on it may learn spurious correlations between political affiliation and predicted veracity. Party-level slice analysis on DistilBERT predictions confirms this concern directly. Republican-affiliated statements achieve a Macro-F1 of 0.2231, substantially below the overall test Macro-F1 of 0.2602, while Democrat-affiliated statements achieve 0.2715, above the overall average. This disparity is documented in the Model Card without softening. NER slice analysis further shows that statements mentioning specific locations (GPE Macro-F1: 0.2011) and organizations (ORG Macro-F1: 0.2068) are consistently harder to classify, suggesting performance varies by the contextual framing of political content as well as party affiliation.

Risk 3: Misuse as a censorship tool. A working misinformation classifier is a dual-use artifact. The same system built for research purposes could in principle be repurposed for automated suppression of political content. This risk is addressed by including explicit intended-use documentation restricting the system to research and fact-checker assistance contexts, and by adding a model license that prohibits commercial or governmental censorship applications. All demo interfaces include prominent disclaimers stating the system is a research prototype only. The party-level performance disparity identified in Risk 2 further underscores why autonomous deployment without human oversight would be harmful regardless of intent.

A full Model Card summarizing the dataset, evaluation results, known limitations, and deployment guidance is available in docs/model_card.md, and the complete Ethics Statement is in docs/ethics_statement.md.

8. REPRODUCIBILITY

The entire pipeline can be reproduced using a single command:

```
bash run.sh
```

All experiments use a fixed random seed of 42. Hyperparameters are logged in experiments/configs/. Training and evaluation metrics are exported to experiments/results/ as JSON files. The environment is captured in requirements.txt. Full training and evaluation completes in approximately 60-75 minutes on a single T4 GPU on Google Colab.

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